

ENTHUSIAST ADVANCE COURSE

PHASE : MED-I (B)

TARGET : PRE-MEDICAL 2024

DATE : 08-10-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	2	3	1	3	3	3	2	4	2	2	4	3	4	3	1	3	2	2	2	1	2	1	2	1	3	2	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	3	4	2	3	3	3	1	1	2	3	2	3	2	1	4	4	1	1	1	2	3	2	2	4	2	2	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	1	1	4	4	3	4	1	4	3	2	2	2	4	1	2	1	3	3	2	1	4	1	4	1	1	3	4	4	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	4	3	1	1	1	3	2	3	4	3	2	4	4	2	3	3	2	2	4	1	1	2	2	1	3	4	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	1	2	4	4	1	2	2	1	1	1	3	3	3	3	4	2	2	2	1	3	3	1	1	3	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	3	1	4	1	2	2	2	3	1	3	3	4	2	2	2	1	3	3	1	3	4	1	3	3	2	3	2	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	1	2	1	4	2	1	3	4	3	3	2	4	2	1	1	3	2	3	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

2. Ans (2)

$$VP^3 = \text{constant} = k \Rightarrow P = \frac{k}{V^{1/3}}$$

$$\text{Also } PV = \mu RT$$

$$\Rightarrow \frac{k}{V^{1/3}} \cdot V = \mu RT \Rightarrow V^{2/3} = \frac{\mu RT}{k}$$

$$\text{Hence } \left(\frac{V_1}{V_2}\right)^{2/3} = \frac{T_1}{T_2}$$

$$\Rightarrow \left(\frac{V}{27V}\right)^{2/3} = \frac{T}{T_2} \Rightarrow T_2 = 9T$$

3. Ans (2)

$$U \propto T$$

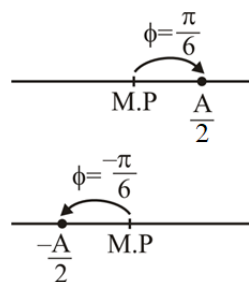
According to given graph

$$P \propto V$$

$$\therefore P \propto T$$

Hence $V = \text{constant}$

7. Ans (3)



$$\Delta\phi = \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$$

9. Ans (2)

$$eV = \frac{hc}{\lambda} - \phi$$

$$\frac{eV}{3} = \frac{hc}{2\lambda} - \phi$$

$$\frac{3hc}{2\lambda} - 3\phi = \frac{hc}{\lambda} - \phi$$

$$\left[\frac{3}{2} - 1\right] \frac{hc}{\lambda} = \frac{2hc}{\lambda_0}$$

$$\frac{1}{2\lambda} = \frac{2}{\lambda_0}$$

$$\boxed{\lambda_0 = 4\lambda}$$

12. Ans (2)

$$K = h\nu - \phi$$

$$2 = h\nu - 2$$

$$h\nu = 4 \text{ eV} \Rightarrow \nu = \frac{4 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$$

$$\nu = 10^{15} \text{ Hz}$$

17. Ans (1)

$$\lambda = \frac{12.27 \text{ \AA}}{\sqrt{V}} = \frac{12.27 \text{ \AA}}{\sqrt{1600}} = 0.3 \text{ \AA}$$

19. Ans (2)

E remain same because $V = \text{constant}$

20. Ans (2)

$$\text{Without copper } C = \frac{\epsilon_0 A}{d}$$

$$\text{with copper } C' = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{\epsilon_r} \right)}$$

$$\text{for copper } \epsilon_r = \infty \text{ and } t = \frac{d}{2}$$

$$\text{Hence } C' = \frac{\epsilon_0 A}{d - \frac{d}{2}} = 2 \frac{\epsilon_0 A}{d}$$

21. Ans (2)

Charge flown through C_3

$$Q_3 = \frac{C_3 (C_1 + C_2) V}{C_1 + C_2 + C_3}$$

Potential difference across C_3

$$\Delta V_3 = \frac{Q_3}{C_3} = \frac{(C_1 + C_2) V}{C_1 + C_2 + C_3}$$

22. Ans (1)

Capacitor behaves as short circuit. So

$$\text{equivalent resistance} = \frac{3R}{2}. \text{ Hence } I = \frac{2V}{3R}$$

23. Ans (2)

Theory based

24. Ans (1)

$$\text{Poynting vector, } S = \frac{P}{4\pi r^2} = \frac{cB_{\max}^2}{2\mu_0}$$

$$\Rightarrow B_{\max} = \sqrt{\frac{2P\mu_0}{4\pi r^2 c}} = \sqrt{\frac{2 \times 15 \times 10^{-7}}{(2)^2 (3 \times 10)^8}} = 5 \times 10^{-8} \text{ T}$$

25. Ans (2)

$$\hat{E} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}} \quad \& \quad \hat{B} = \frac{-\hat{i} - \hat{j}}{\sqrt{2}}$$

$$\hat{E} \times \hat{B} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}} \times \frac{-\hat{i} - \hat{j}}{\sqrt{2}}$$

$$= \frac{2\hat{k}}{2}$$

$$= \hat{k}$$

26. Ans (1)

using KCL, $I_3 = I_1 + I_2$

$$I_3 = 3A \sin \omega t + 4A \cos \omega t$$

$$I_3 = R \sin (\omega t + \theta)$$

$$\text{where } R = \sqrt{(3A)^2 + (4A)^2} = 5A$$

$$\tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$$

27. Ans (3)

$$E = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= \sqrt{(80)^2 + (40 - 100)^2}$$

$$= \sqrt{6400 + 3600} = \sqrt{10000}$$

$$= 100V$$

28. Ans (2)

$$V = 50 \times 2 \sin(100\pi t) \cos(100\pi t)$$

$$= 50 \sin(200\pi t)$$

$$V_0 = 50 \text{ and } \omega = 200 \pi$$

$$f = \frac{\omega}{2\pi} = \frac{200\pi}{2\pi} = 100 \text{ Hz}$$

29. Ans (3)

$$i_{\text{rms}} = \sqrt{\frac{i_1^2 + i_2^2}{2}} = \frac{1}{\sqrt{2}} (i_1^2 + i_2^2)^{1/2}$$

30. Ans (3)

$$\cos \phi = \frac{R}{Z}$$

$$\cos \phi = \frac{10}{20}$$

$$\cos \phi = \frac{1}{2}$$

$$\phi = 60^\circ$$

31. Ans (3)

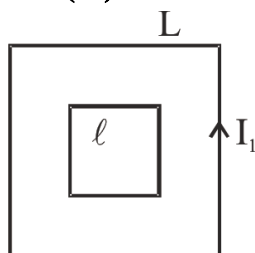
$$X_L = 2\pi fL$$

$$X_L \propto f \text{ (graph straight line)}$$

$$X_C = \frac{1}{2\pi fC}; \quad X_C \propto \frac{1}{C}$$

(graph rectangular hyperbola)

32. Ans (2)



$$M = \frac{\phi_2}{I_1} = \frac{N_2 A_2 B_1}{I_1}$$

$$= \frac{1 \times l^2 \times 2\sqrt{2} \mu_0 I_1}{I_1 (\pi L)}$$

$$M \propto \frac{l^2}{L}$$

33. Ans (3)

Output power = 140 watt

Input power = $240 \times 0.7 = 168$ watt

$$\text{Efficiency} = \frac{\text{output power}}{\text{input power}} \times 100$$

$$= \frac{140}{168} \times 100 = 83.3\%$$

34. Ans (4)

Transformer does not work on D.C.

35. Ans (2)

Theoretical fact.

SECTION-B

36. Ans (3)

$$Q_1 = 10 \times 1 \times 10 = 100 \text{ cal}$$

$$Q_2 = 10 \times 0.5[0 - (-20)] + 10 \times 80$$

$$= (100 + 800) \text{ cal} = 900 \text{ cal}$$

As $Q_1 < Q_2$, so ice will not completely melt and final temperature = 0°C

As heat given by water in cooling up to 0°C is only just sufficient to increase the temperature of ice from -20°C to 0°C , hence mixture in equilibrium will consist of 10 gm of ice and 10 gm of water, both at 0°C .

37. Ans (3)

The efficiency of carnot engine,

$$\eta = 1 - \frac{T_2}{T_1}$$

where, T_2 is temperature of sink and T_1 is temperature of hot reservoir.

According to question,

$$\frac{1}{5} = 1 - \frac{T_2}{T_1} \quad \dots\dots(i)$$

$$\text{And, } \frac{1}{3} = 1 - \frac{T_2 - 50}{T_1} \quad \dots\dots(ii)$$

From eq. (i), we get

$$\frac{T_2}{T_1} = \frac{4}{5} \Rightarrow T_1 = \frac{5}{4} T_2$$

Substituting value of T_1 in eq. (ii), we get

$$\frac{1}{3} = 1 - \frac{T_2 - 50}{\frac{5}{4} T_2}$$

$$\text{or, } \frac{4(T_2 - 50)}{5T_2} = \frac{2}{3}$$

$$\text{or, } T_2 - 50 = \frac{2}{3} \times \frac{5}{4} T_2 \text{ or } T_2 - \frac{5}{6} T_2 = 50$$

$$\therefore T_2 = 50 \times 6 = 300 \text{ K}$$

38. Ans (3)

Slop of curve $\frac{dT}{dt} \Rightarrow$ Rate of cooling

$$|(\text{slope})_x| > |(\text{slope})_y|$$

(Rate of cooling) $\Rightarrow (R_f)_x > (R_f)_y$

So $[e_x > e_y]$

From Kirchhoff $e \propto a$

$$[a_x > a_y]$$

41. Ans (2)

no of photons = $n \times t$

$$= 5 \times 10^{24} P \times \lambda \times t$$

$$= 5 \times 10^{24} E \times \lambda$$

$$= 5 \times 10^{24} \times 10^{-7} \times 5 \times 10^{-7}$$

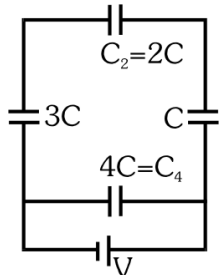
$$= 2.5 \times 10^{11}$$

42. Ans (3)

$$\lambda = \frac{h}{\sqrt{2mqv}} \Rightarrow \lambda \propto \frac{1}{\sqrt{mq}}$$

$$\lambda_e > \lambda_p > \lambda_\alpha$$

44. Ans (3)



$$Q_4 = 4CV$$

$$Q_2 = \left(\frac{6}{11} C \right) V = \frac{6CV}{11}$$

$$\Rightarrow \frac{Q_2}{Q_4} = \frac{6CV}{11} \times \frac{1}{4CV} = \frac{3}{22}$$

45. Ans (2)

$$\Delta Q = 2CV - (-CV) = 3CV$$

$$W_B = \Delta Q(2V) = 6CV^2$$

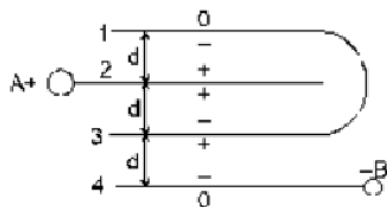
$$\Delta U = U_f - U_i = \frac{1}{2} C(2V)^2 - \frac{1}{2} CV^2 = \frac{3CV^2}{2}$$

$$\therefore \text{Heat} = W_B - \Delta U = \frac{9CV^2}{2}$$

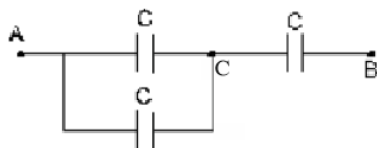
$$U_f = \frac{1}{2} C(2V)^2 = 2CV^2$$

$$\therefore \frac{\text{Heat}}{U_f} = \frac{9}{4} = 2.25$$

46. Ans (1)



The modified circuit is



$$C_{eq} = \frac{2C}{3}$$

47. Ans (4)

$$E = B \times C = 2 \times 10^{-7} \times 3 \times 10^8 = 60 \text{ V/m}$$

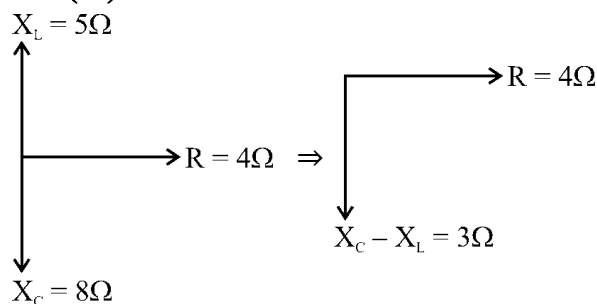
$$\vec{C} \perp \vec{E}, \vec{B}$$

$$\text{So, } \vec{E}_y = 60 \sin(0.5 \times 10^3 x - 1.5 \times 10^8 t) \text{ V/m}$$

48. Ans (4)

$$\begin{aligned} i_d &= \epsilon_0 \frac{d\phi_E}{dt} \\ &= \epsilon_0 A \frac{dE}{dt} \\ &= \frac{10^{-9}}{9 \times 4\pi} \times \pi \times (1.2)^2 \times \frac{200}{5} \\ &= 1.6 \times 10^{-9} \text{ Amp} \end{aligned}$$

49. Ans (1)



$$\begin{aligned} \tan \phi &= \frac{X_C - X_L}{R} = \frac{3}{4} \\ \phi &= \tan^{-1} \left(\frac{3}{4} \right) \end{aligned}$$

$X_C > X_L$ so current lead voltage.

50. Ans (1)

$$E \cdot 2\pi r = \pi a^2 \frac{dB}{dt}$$

SUBJECT : CHEMISTRY

SECTION-A

51. Ans (1)

$$[\text{Ni}(\text{CN})_4]^{-2}, [\text{Ni}(\text{CO})_4] \text{ no unpaired } e^-$$

$$\mu_s = 0$$

52. Ans (2)

$$[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$$

C_2H_4 = neutral ligand

$$x + 0 + 3 - 1 = -1$$

$$x = -1 + 3 \Rightarrow x = +2 \text{ o.s.} = +2$$

coordination number = 4

53. Ans (3)

$$\text{CFSE} = -0.4 \Delta_0 n_{t_2g} + 0.6 \Delta_0 n_{e_g}$$

$$[\text{Ni}(\text{H}_2\text{O})_6]^{+2} \text{ has CFSE} = -1.2 \Delta_0$$

54. Ans (2)

NO_2^- Ambidentate Ligand

Complexes mentioned are linkage isomer

55. Ans (2)

$\Rightarrow [\text{CrBr}_2(\text{H}_2\text{O})(\text{NH}_3)_3]$; No. of stereoisomers = 3,
cis = 1 trans = 2.

$\Rightarrow [\text{Zn}(\text{gly})_2]$; No. of stereoisomers = 2, both are in
d and l forms.

$\Rightarrow [\text{PtBrCl}(\text{CN})(\text{H}_2\text{O})^-]$; No. of stereoisomers = 3,
all are geometrical isomers.

$\Rightarrow [\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$; No. of stereoisomers = 0.

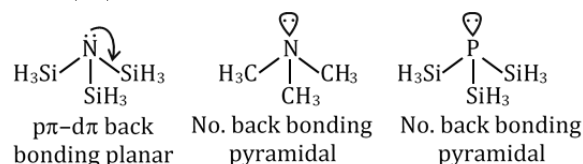
56. Ans (4)

$[\text{Ma}_3\text{b}_3]$ show facial & meridional isomer.

57. Ans (2)

Oxalate is a bidentate ligand.

59. Ans (2)



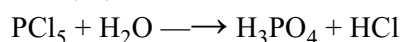
60. Ans (4)

NF_5 [N belongs to 2nd period so it does not
have d-orbitals]

BiI_5 [Bi is in +5 O.S $\rightarrow$ due to inert pair effect
 Bi^{+5} is not stable]

PbI_4 [Pb is in +4 O.S $\rightarrow$ due to inert pair effect
 Pb^{+4} is not stable]

61. Ans (3)



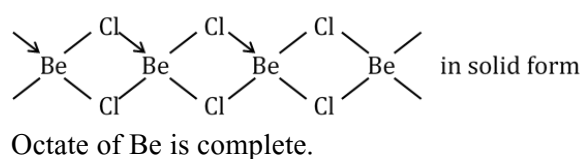
62. Ans (1)

Conc. H_2SO_4 is dehydrating agent

63. Ans (1)

sheet silicate

64. Ans (4)



73. Ans (2)

$$\begin{aligned} K_w &= [\text{H}_3\text{O}^+][\text{OH}^-] = [\text{H}_3\text{O}^+]^2 \\ &= (10^{-6})^2 = 10^{-12} \end{aligned}$$

75. Ans (1)

TC: @NEETxNOAH

$$\begin{aligned} \text{eq of } 0.2 \text{ M } \text{H}_2\text{SO}_4 &= \frac{2 \times 0.2\text{M}}{1000} \times 100 \\ &= 0.04 \text{ eq} \end{aligned}$$

$$\begin{aligned} \text{eq of } .2 \text{ M } \text{NaOH} &= \frac{0.2}{1000} \times 100 \\ &= 0.02 \text{ eq} \end{aligned}$$

$$\text{left } [\text{H}^+] = .04 - .02 = .02$$

$$\text{Total volume} = 200$$

$$\begin{aligned} \Rightarrow [\text{H}^+]_{\text{final}} &= \frac{0.02}{200} = .0001 \\ &= 10^{-4} \end{aligned}$$

$$\text{pH} = 4$$

SUBJECT : BIOLOGY-I

SECTION-A

105. Ans (4)

NCERT Pg. # 166

110. Ans (2)

NCERT XI Pg # 219

113. Ans (1)

NCERT-XI Pg. # 212

114. Ans (2)

NCERT Pg. # 211

115. Ans (2)

NCERT Pg. # 164

116. Ans (1)

NCERT Pg. # 159

122. Ans (3)

NCERT (XII) Pg. # 211

123. Ans (4)

NCERT (XII) Pg. # 211, para-12.2.2

125. Ans (2)

NCERT Page # 197

127. Ans (4)

NCERT XIIth Pg.#199

129. Ans (2)

NCERT XII, Pg.#201

132. Ans (1)

NCERT, Pg.# 212

133. **Ans (1)**
 NCERT XII Pg # 200(E), 220(H)

SECTION-B

141. **Ans (2)**
 NCERT Pg. # 163
142. **Ans (1)**
 NCERT Pg. # 164
145. **Ans (1)**
 NCERT–XII, Pg. # 121
146. **Ans (1)**
 NCERT-XII Pg. # 195
148. **Ans (2)**
 NCERT (XIIth) Pg. # 198

SUBJECT : BIOLOGY-II
SECTION-A

152. **Ans (1)**
 NCERT (XII) Pg. # 202,203 para-11.3.3
155. **Ans (4)**
 NCERT (XII) Pg. # 212
156. **Ans (1)**
 NCERT Pg. # 174
158. **Ans (2)**
 NCERT Pg. # 177, para 9.4
160. **Ans (3)**
 NCERT (XII) Pg. # 176

161. **Ans (1)**
 NCERT Pg. # 176
170. **Ans (3)**
 NCERT (XII) Pg. # 99(E), 108(H)
171. **Ans (1)**
 NCERT (XII) Pg # 107, Fig. 6.8
174. **Ans (1)**
 NCERT Pg. # 321
175. **Ans (3)**
 NCERT Pg. # 321
177. **Ans (2)**
 NCERT, Pg.# 323, 21.6.1

SECTION-B

187. **Ans (1)**
 NCERT Page # 209
188. **Ans (3)**
 NCERT XII, Pg. # 210
189. **Ans (4)**
 NCERT-Pg. # 177 Para-4
194. **Ans (2)**
 NCERT Page-106
195. **Ans (1)**
 NCERT XII Page No. # 318
196. **Ans (1)**
 NCERT Pg. # 318 (21.3.1)